

PHYSICS II MIDTERM 1

1. a) $x = 8.00$ $q = 1.00 \mu C$

$$F = k \frac{|q_1 q_2|}{r^2}$$

$+q @ x = 0 \text{ cm}$
 $-2q @ x = 10 \text{ cm}$
 $+q @ x = 15 \text{ cm}$

~~$= 44.5 \text{ N}$~~
 $= 44.57 \text{ N}$ to right

$$F_1 = k \frac{(1.00 \times 10^{-6})(1.00 \times 10^{-6})}{(0.08)^2}$$

b) $2.65 \times 10^8 \text{ N/C}$ to left

$$F_{\text{net}} = F_1 + F_2 - F_3$$

$$E = k \frac{|q|}{r^2}$$

$$E_1 = 1.49 \times 10^6 \text{ N/C}$$

$$E_2 = 2.50 \times 10^6 \text{ N/C}$$

$$E_3 = 2.30 \times 10^6 \text{ N/C}$$

$$E_{\text{net}} = E_1 + E_3 - (E_2 + E_3)$$

2. a) $q_a = q_b = +7.50 \mu C$ @ top left & top right
 $q_c = q_d = -7.50 \mu C$ @ bottom left & bottom right
 Charge $q = 24.00 \mu C$ is located @ center

force on q is downward from vertical axis

b) $r = \frac{10.0}{\sqrt{2}} = 7.07 \text{ cm} = 0.0707 \text{ m}$

$$F = k \frac{|q_1 q_2|}{r^2} = 27.01$$

$$F_{\text{net}} = 2 \cdot F_v + 2 \cdot F_v$$

$$= 76.39 \text{ N downward}$$

3.

$$q v = \frac{1}{2} m v^2$$

$$q = 1.602 \cdot 10^{-19}$$

$$V = 401 \text{ kV} = 401 \cdot 10^3 \text{ V}$$

$$m = 9.109 \cdot 10^{-31}$$

$$v = \sqrt{\frac{2qV}{m}} \rightarrow 1.19 \times 10^8 \text{ m/s}$$

4.

$$\mathcal{E} = \frac{\Delta V}{d} \rightarrow \Delta V = \mathcal{E} \cdot d$$

$$\mathcal{E} = 7.50 \cdot 10^4 \text{ V/m}$$

$$d = 4.00 \text{ cm} = .0400 \text{ m}$$

$$\Delta V = 3000 \text{ V}$$

$$b) V = \mathcal{E} \cdot x$$

$$x = 1.00 \text{ cm} = .0100 \text{ m}$$

$$\mathcal{E} = 7.50 \cdot 10^4 \text{ V/m}$$

$$= 750 \text{ V @ } 1.00 \text{ cm}$$

Zero plate

$$3000 - 750 = 2250 \text{ V}$$

c)

$$\mathcal{E} = \frac{\Delta V}{d}$$

$$\Delta V = 80.0 \text{ mV} = .0800 \text{ V}$$

$$d = 9.00 \text{ nm} = 9.00 \cdot 10^{-9} \text{ m}$$

$$= 8.89 \times 10^6 \text{ V/m}$$

5.

$$U = qv$$

$$U = 32.0 \text{ keV} = 32.0 \cdot 10^3 \text{ eV}$$

$$q = 2e = 2(1.602 \cdot 10^{-19}) \text{ C}$$

$$v = 16,000$$

$$d = 2.00 \text{ cm}$$

$$= 8.00 \times 10^5 \text{ V/m}$$

$$v = \frac{U}{q}$$

$$= 16,000 \text{ V or } 16.0 \text{ kV}$$

$$6. \quad k_i + U_i = k_f + U_f$$

$$k = U$$

$$U = k \frac{q_1 q_2}{r}$$

$$1k = 5.00 \times 10^6 \cdot (1.602 \cdot 10^{-19})^2 \text{ J}$$

$$= 45.6 \text{ fm}$$

$$4.56 \times 10^{-14}$$

$$1k = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

$$q_1 = 2e = 2(1.602 \times 10^{-19}) \text{ C}$$

$$q_2 = 79e = 79(1.602 \cdot 10^{-19})$$

$$r = k \frac{q_1 q_2}{1k}$$

Unit 1: Section 2

$$1) \quad C = \frac{Q}{V}$$

$$Q = 3.00 \mu\text{C} = 3.00 \cdot 10^{-6} \text{ C}$$

$$V = 120 \text{ V}$$

$$= 2.50 \times 10^{-8} \text{ F} \text{ or } 25 \text{ nF}$$

$$2. \quad a) \quad U = \frac{1}{2} C V^2$$

$$C = 10 \mu\text{F} = 10 \cdot 10^{-6} \text{ F}$$

$$V = 9 \times 10^3 \text{ V}$$

$$\text{Energy in } 10 \mu\text{F capacitor} = 405 \text{ J}$$

$$b) \quad V = \sqrt{\frac{2U}{C}}$$

$$U = 40 \text{ J}$$

$$C = 8 \mu\text{F}$$

$$= 3.16 \text{ kV} \text{ or } 3162 \text{ V}$$

$$3). \quad a) \quad C_{\text{total}} = 4C$$

$$C_{\text{total}} = 10 \mu\text{F}$$

$$10 \cdot 10^{-6} = 4C$$

$$= 2.50 \mu\text{F}$$

Charge on each cap.

$$C = 2.50 \mu\text{F} = 2.50 \cdot 10^{-6} \text{ F}$$

$$V = 9 \cdot 10^3 \text{ V}$$

$$= 22.5 \text{ mC}$$

$$0.225 \text{ C of charge}$$

4.

$$V = IR$$

$$I = \frac{V}{R_{total}}$$

$$R_{total} = R_{resistor} + R_{LED}$$

$$R_{res} = 1.00 k\Omega = 1000 \Omega$$

$$R_{LED} = 3.00 \Omega$$

$$V = 3.0 V$$

$$a) = 2.99 \text{ mA}$$

$$b) P = I^2 R$$

$$I = 2.99 \text{ mA}$$

$$R = 3.00 \Omega$$

$$= 26.8 \mu W$$

Unit 2 section 3:

1.

$$a) \frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$R_1 = 10 k\Omega$$

$$R_2 = 5 k\Omega$$

$$R_{tot} = 2 k\Omega$$

$$R_3 = 3.00 k\Omega$$

$$b) I = \frac{V}{R_{tot}}$$

$$V = 12 V$$

$$R_{tot} = 2 k\Omega = 2000 \Omega$$

$$= 6.00 \text{ mA}$$

$$c) I_n = \frac{V}{R_n}$$

$$V = 16 V$$

$$I_1 = 1.60 \text{ mA}$$

$$I_2 = 2.40 \text{ mA}$$

$$I_3 = 2.40 \text{ mA}$$

$$I_1 + I_2 + I_3 = 6.00 \text{ mA}$$

2. a) $I = \frac{V}{R}$ $V = 1.5V$
 parallel $V = 3V$
 $6.00 \text{ mA} \Rightarrow$ parallel circuit
 $P = I^2 R = 18.0 \text{ mW}$

b) $R_{\text{total}} = R + r_1 + r_2$ $r_1 = .5$
 $= 3.99 \text{ mA}$ $r_2 = .5$
 $R = 500$ } - series
 Power P dissipated in resistor
 $P = I^2 R \rightarrow 17.93 \text{ mW}$

Parallel: New Current: 3.997 mA
 New power: 17.98

$r_{\text{eq}} = \frac{r}{\text{# of parallel}} = \frac{.5}{2}$ $r_{\text{tot, parallel}} = K r_{\text{eq}}$
 $= 3.997 \text{ mA}$

3. $I = \frac{V_{\text{tot}}}{R_{\text{tot}}}$ $3 \text{ cells} = 1.58V$
 $1 \text{ cell dry cell} = 1.53V$

$V_{\text{tot}} = (3 \cdot 1.58) + 1.53$

$r_{\text{load}} = 10.00$ $I = \text{current flowing through circuit}$
 $3 \cdot \frac{102}{400} = .617 \text{ A}$

$R_{\text{tot}} = R_{\text{load}} + (3 \cdot .02) + .10$

$n = I^2 R$

b) $I = .617 \text{ A} \Rightarrow 3.81 \text{ W}$
 $R_{\text{load}} = 10.00 \Omega$

c. $P = I^2 R$

$$I = \sqrt{\frac{P}{R}}$$

New current when power drops

$.500 \text{ W is } .224 \text{ A}$

$$R_{\text{tot, new}} = \frac{V_{\text{tot}}}{I_{\text{new}}}$$

$\Rightarrow$ only $.500 \text{ W}$ to be supplied

$\rightarrow 18.04 \Omega$

Unit 3: section 4

1) $F = qv \cdot B$

$i = \text{right direction}$

$j = \text{up}$

$k = \text{out of page}$

CASE	V Direction	B Direction	F direction
a	down	out of page	Right hand rule positive left negative
b	Right	Up	Right hand rule out of page & positive negative into page
c	Right	Into page	Right hand rule up negative charge down

2) $F = qvB \sin \theta$

$\theta = \arcsin\left(\frac{F}{qvB}\right)$

$F = 1.40 \times 10^{-10} \text{ N}$

$q = 1.602 \times 10^{-19} \text{ C}$

$v = 4 \times 10^3 \text{ m/s}$

$B = 1.25 \text{ T}$

$\theta = 180^\circ - \theta$

$= 16.07^\circ \Rightarrow 169.93^\circ$

3. a) $qVB = \frac{mv^2}{r}$

$q = \frac{mv}{Bv}$

$= 4.80 \times 10^{-16} \text{ C}$

$m = 2.66 \times 10^{-26} \text{ kg}$

$v = 3 \times 10^6 \text{ m/s}$

$B = 1.20 \text{ T}$

$R = .671 \text{ m}$

b. $e = 1.602 \cdot 10^{-19} \text{ C}$

$$\frac{q_{O-16}}{e}$$

= O^{-16} ion 2 electron charge is 3.00

which means O ion has +3e charge

c) $\frac{m_{O-16}}{m_{O-18}} = \frac{16}{18}$

$$r_{O-18} = r_{O-16} \cdot \frac{m_{O-18}}{m_{O-16}}$$

$$= \underline{.260 \text{ m}}$$